

2.10.52

EE25BTECH11023 - Venkata Sai

Question:

Let $\mathbf{a} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} + \hat{\mathbf{k}}$, $\mathbf{b} = \hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$ and $\mathbf{c} = \hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}}$. A vector in the plane of $\mathbf{a}$ and $\mathbf{b}$ whose projection on $\mathbf{c}$ is $\frac{1}{\sqrt{3}}$, is

Solution:

Given

$$\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \mathbf{b} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \mathbf{c} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}. \quad (1)$$

Let $\mathbf{v}$ be the vector in the plane of $\mathbf{a}$ and $\mathbf{b}$

$$[\mathbf{v} \quad \mathbf{a} \quad \mathbf{b}] = 0 \implies \mathbf{v} = \alpha\mathbf{a} + \beta\mathbf{b} \quad (2)$$

$$\mathbf{v} = \alpha \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} \alpha + \beta \\ 2\alpha - \beta \\ \alpha + \beta \end{pmatrix}. \quad (3)$$

Mathematically, the projection of $\mathbf{v}$ on $\mathbf{c}$ is defined as

$$\mathbf{d} = k\mathbf{c}, \text{ such that } (\mathbf{v} - \mathbf{d})^\top \mathbf{d} = 0 \quad (4)$$

yielding

$$(\mathbf{v} - k\mathbf{c})^\top \mathbf{d} = 0 \quad (5)$$

$$\text{or, } k = \frac{\mathbf{v}^\top \mathbf{c}}{\|\mathbf{c}\|^2}, \implies \mathbf{d} = \frac{\mathbf{v}^\top \mathbf{c}}{\|\mathbf{c}\|^2} \mathbf{c}. \quad (6)$$

$$\|\mathbf{c}\|^2 = \mathbf{c}^\top \mathbf{c} = \begin{pmatrix} 1 & 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \quad (7)$$

$$= 1 + 1 + 1 = 3 \implies \|\mathbf{c}\| = \sqrt{3} \quad (8)$$

$$\mathbf{v}^\top \mathbf{c} = \begin{pmatrix} \alpha + \beta & 2\alpha - \beta & \alpha + \beta \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \quad (9)$$

$$= \alpha + \beta + 2\alpha - \beta - \alpha - \beta = 2\alpha - \beta \quad (10)$$

$$\mathbf{d} = \frac{2\alpha - \beta}{3} \mathbf{c} \quad (11)$$

Given $\|\mathbf{d}\| = \frac{1}{\sqrt{3}}$, Then

$$\frac{|2\alpha - \beta|}{3} \|\mathbf{c}\| = \frac{|2\alpha - \beta|}{3} \sqrt{3} = \frac{1}{\sqrt{3}} \implies |2\alpha - \beta| = 1 \quad (12)$$

From equation(3) we know that that x and z coordinates in our required vector must be equal.

If $2\alpha - \beta = -1$, from the most reliable option i.e option(1) which has the same x and z coordinates, we get

$$\alpha + \beta = 4 \text{ and } 2\alpha - \beta = -1 \quad (13)$$

On adding gives

$$3\alpha = 3 \implies \alpha = 1 \text{ and } \beta = 3 \quad (14)$$

$$\mathbf{v} = \begin{pmatrix} 4 \\ -1 \\ 4 \end{pmatrix} \quad (15)$$

If $2\alpha - \beta = 1$, from the options x and z coordinates are not equal, No option satisfy.

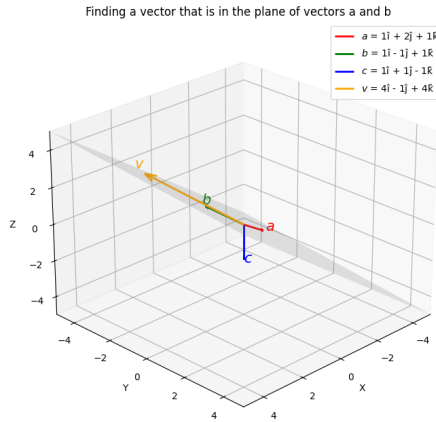


Fig. 0.1